

VALVE POWER BOARD

RUBY - REV A



RESPONSIBLE ENGINEER:
NICHOLAS ALVES

REVIEWERS:

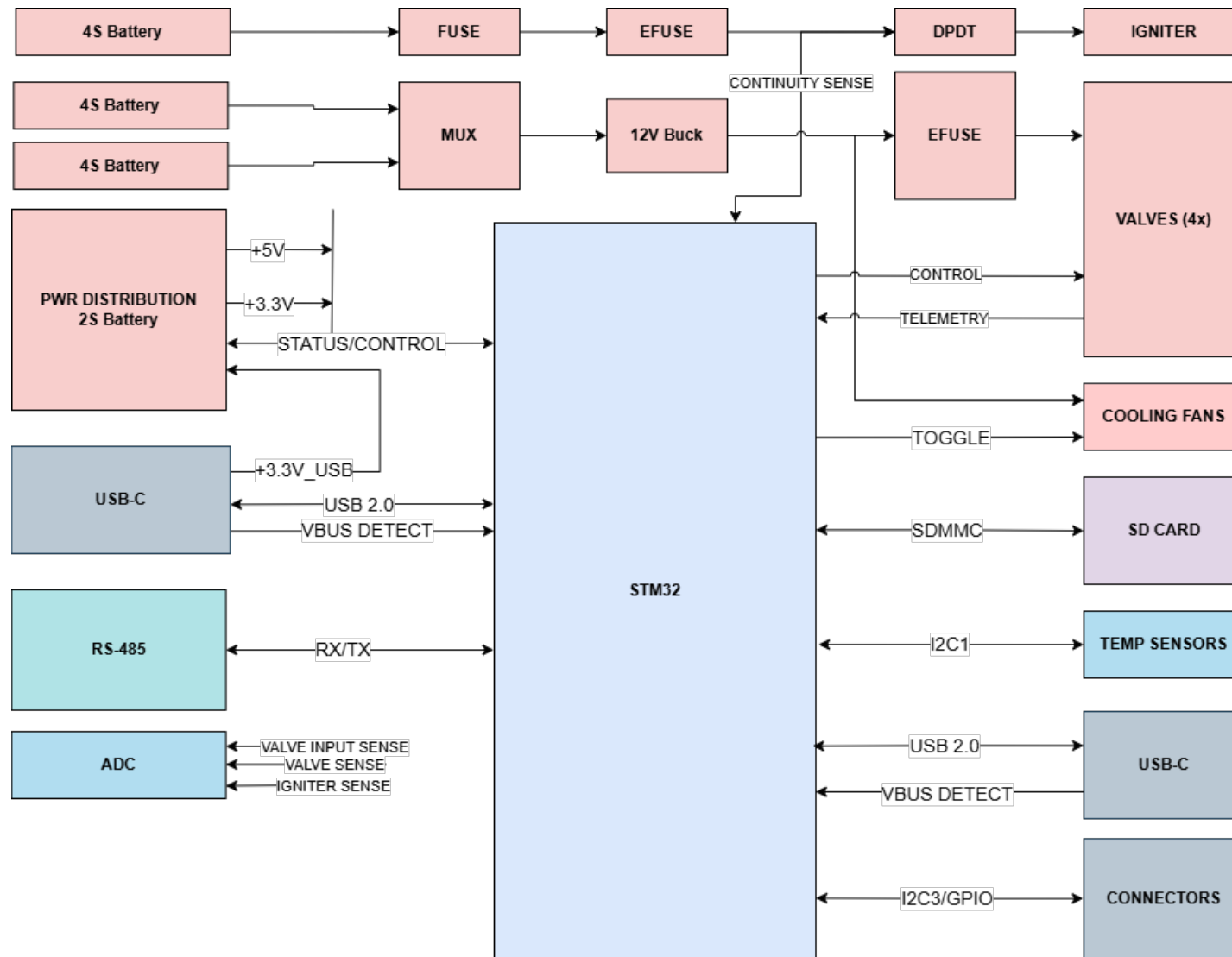
SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/
File: top-level.kicad_sch

Title: TITLE PAGE

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

Rev: A
Id: 1/32



Sheet: /Block Diagrams/
File: block_diagrams.kicad_sch

Title: BLOCK DIAGRAM

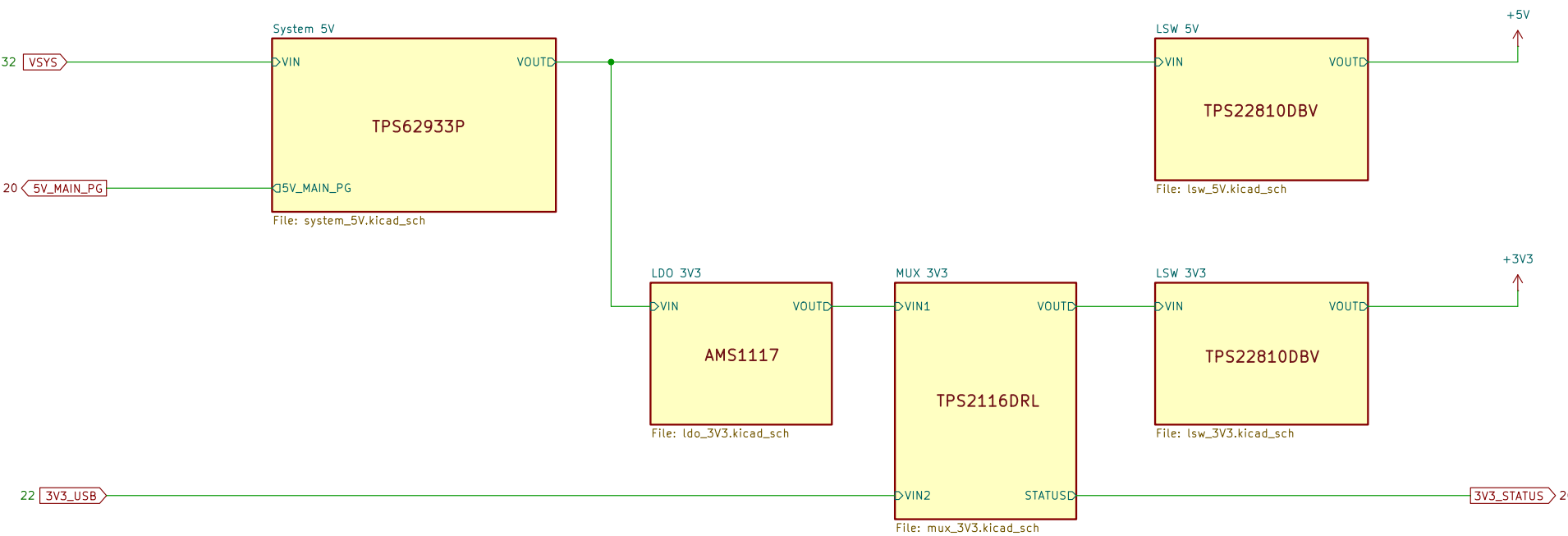
Size: A4 Date: 2026-10-08

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Id: 2/32

SYSTEM POWER



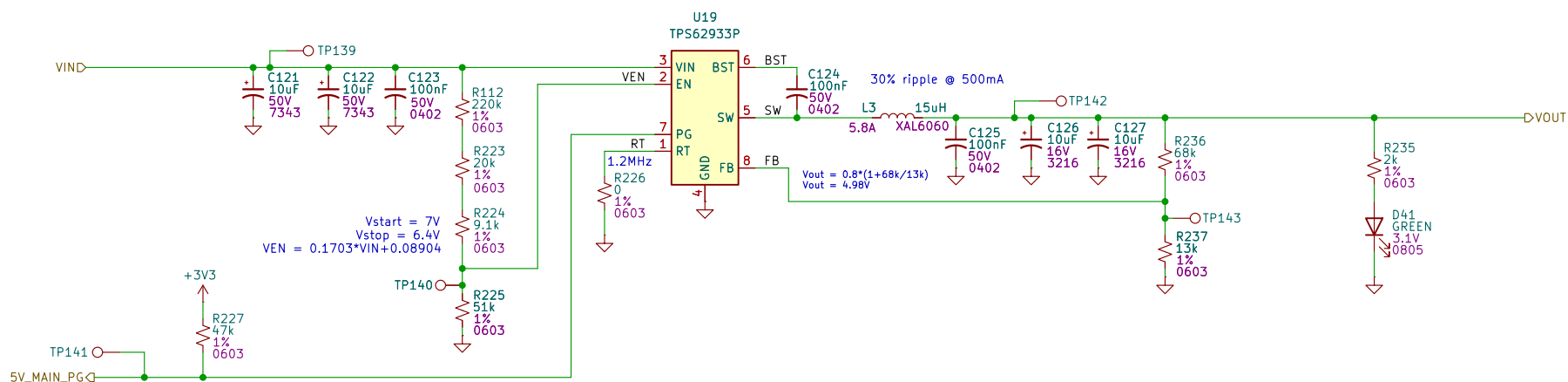


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Sheet: /System Power/
File: system_power.kicad_sch

Title: SYSTEM POWER

Size: A4	Date: 2026-10-08	Rev: A
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Sheet: /System Power/System 5V/
File: system_5V.kicad_sch

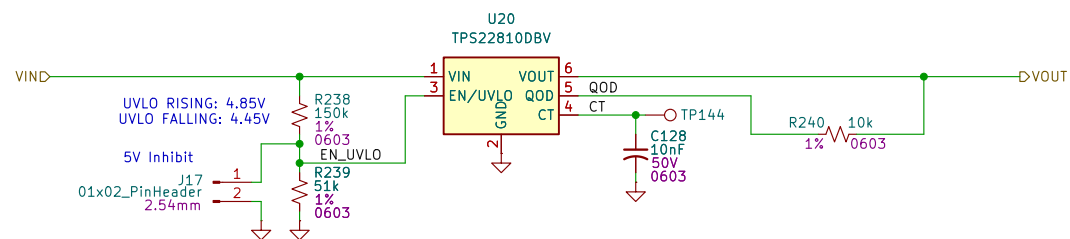
Title: SYSTEM 5V

Size: A4 Date: 2026-10-08

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Sheet: /System Power/LSW 5V/
File: lsw_5V.kicad_sch

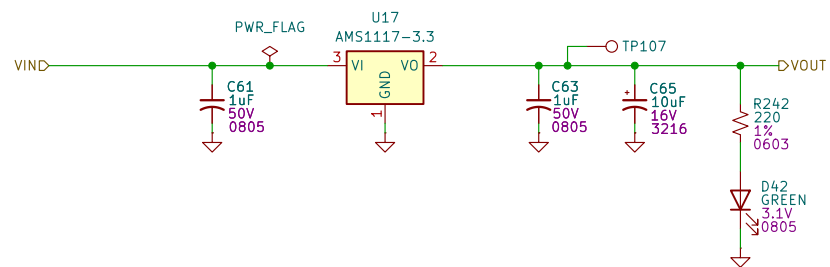
Title: LSW 5V

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 5/32



Sheet: /System Power/LDO 3V3/
File: ldo_3V3.kicad_sch

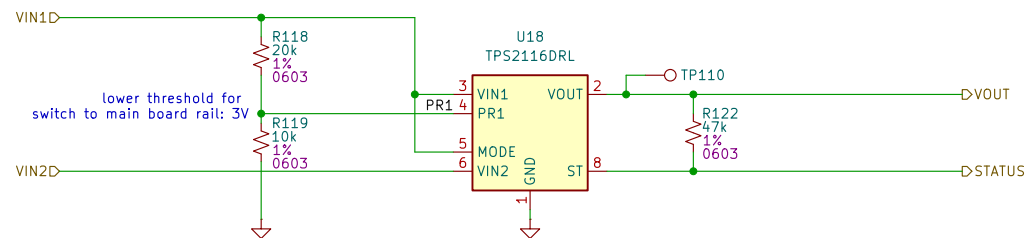
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Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

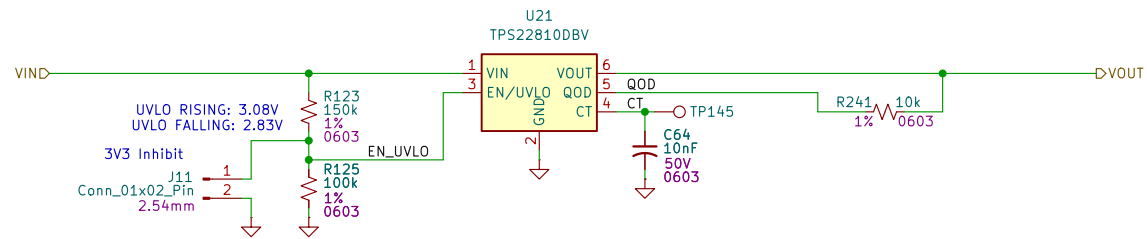
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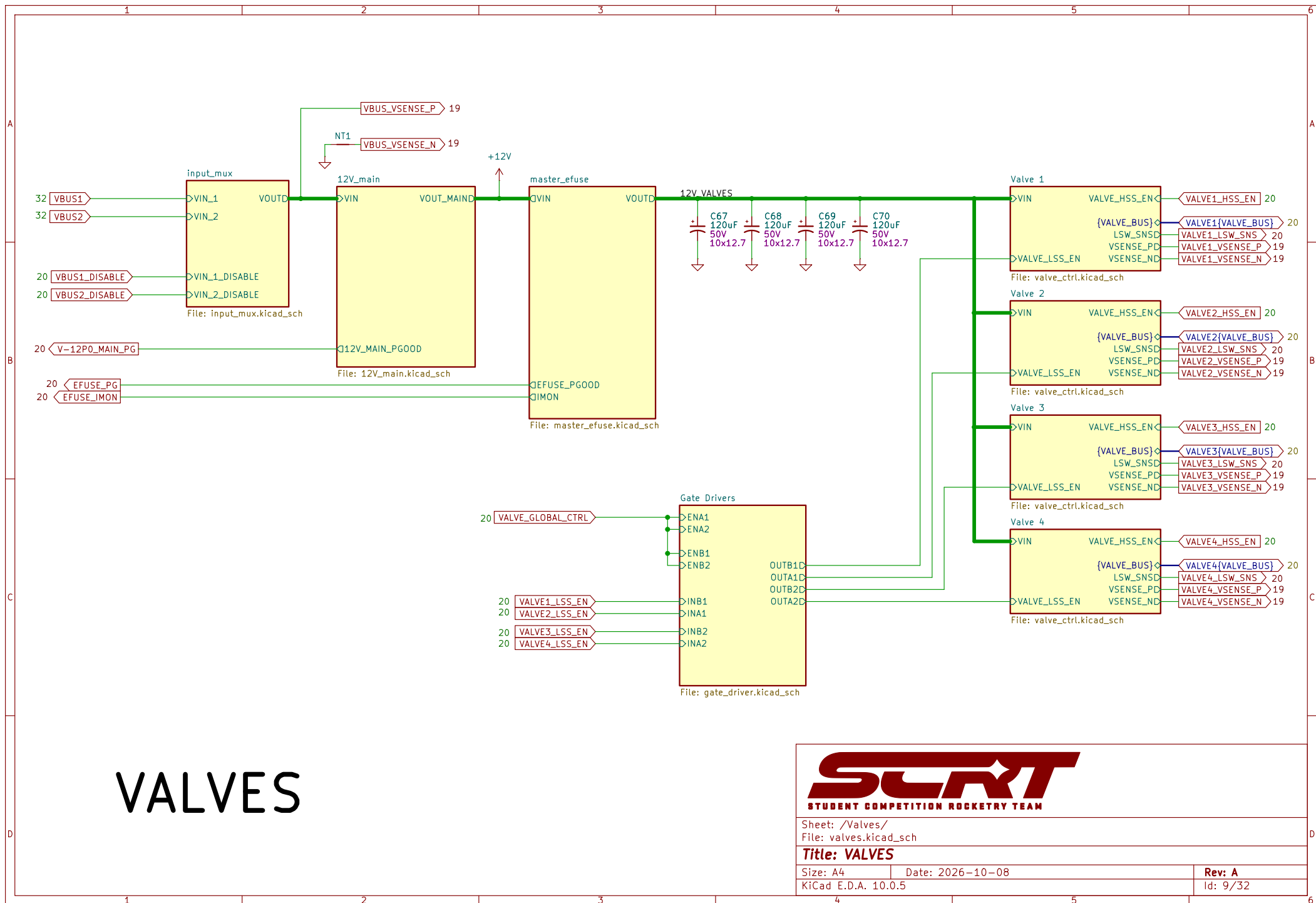
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File: mux_3V3.kicad_sch

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Size: A4	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.5		Id: 7/32

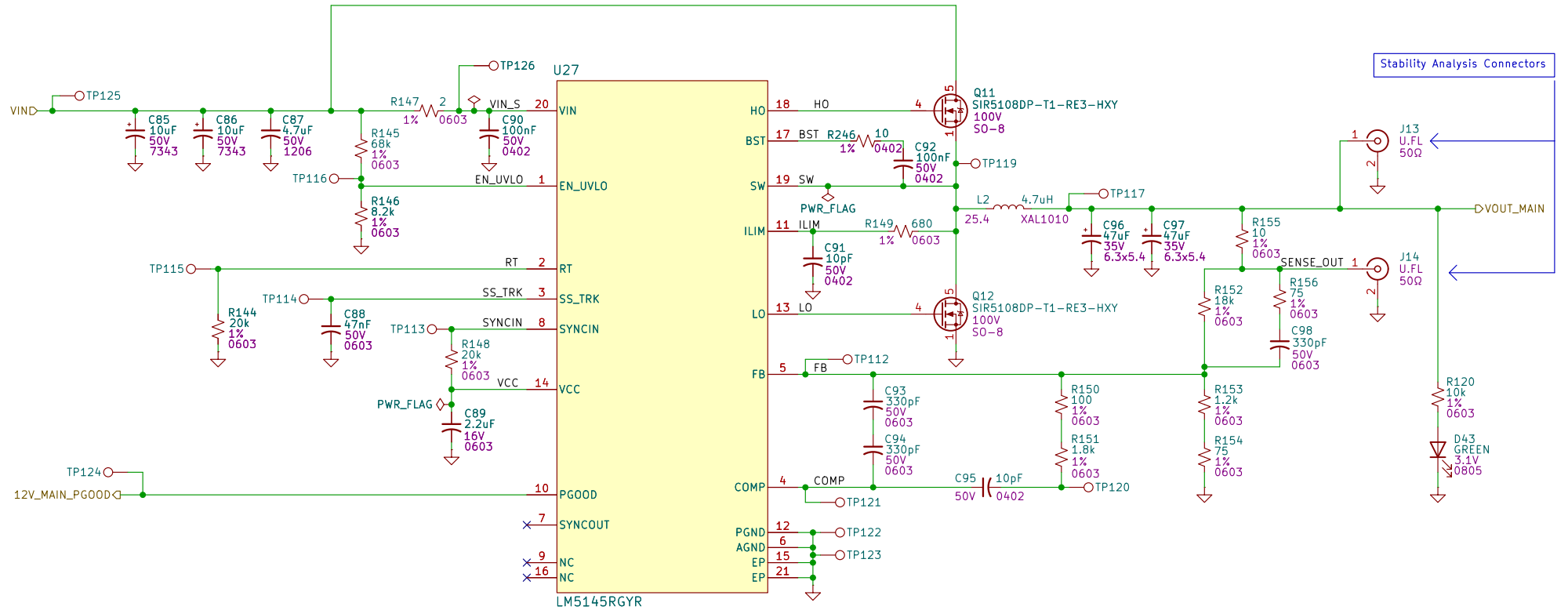


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Title: LSW 3V3		
Size: A4	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.5		Id: 8/32





12V MAIN



Sheet: /Valves/12V_main/
File: 12V_main.kicad_sch

Title: 12V MAIN

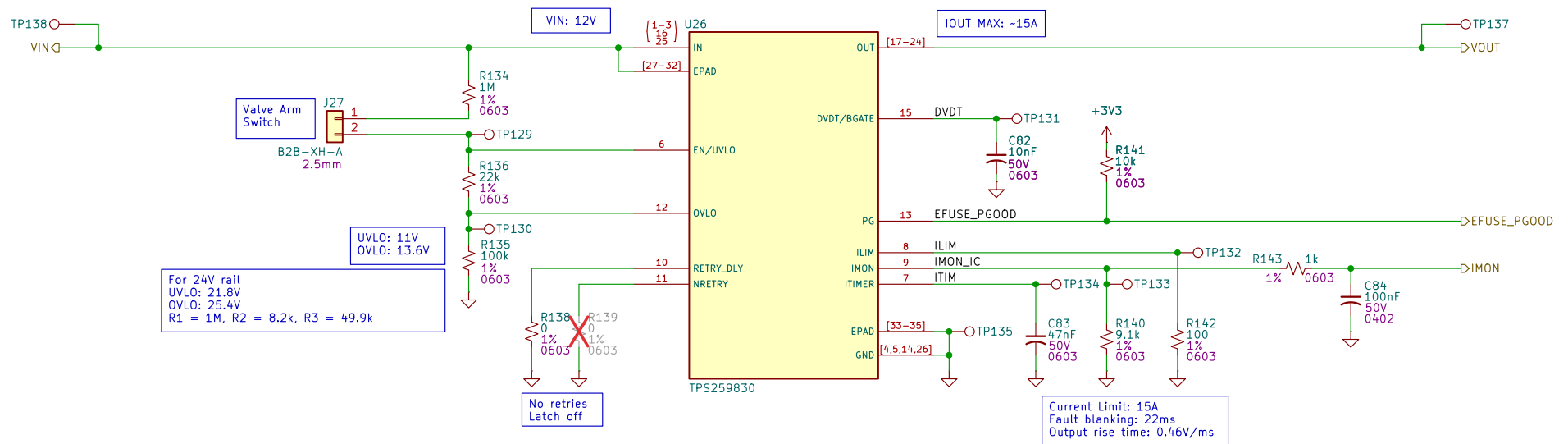
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KiCad E.D.A. 10.0.5

Rev: A

Id: 11/32

EFUSE

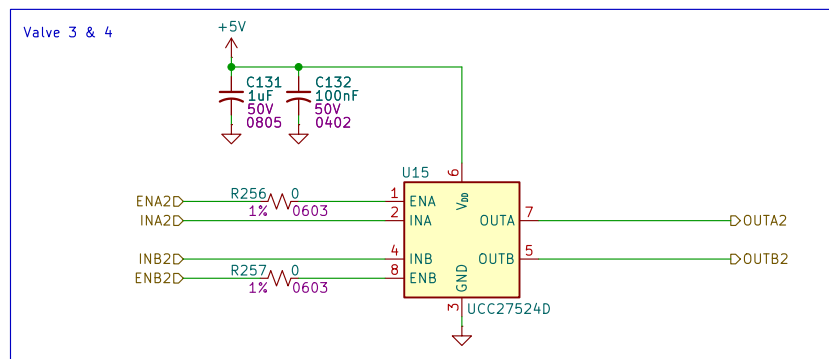
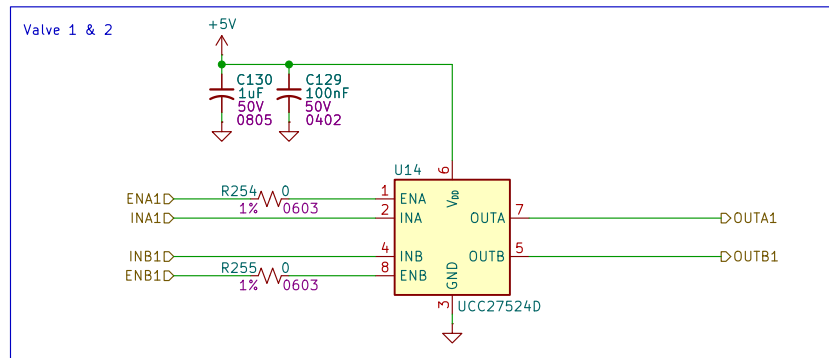


Sheet: /Valves/master_efuse/
File: master_efuse.kicad_sch

Title: EFUSE

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

Rev: A
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Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



Title: Valve Output

Id: 14/32

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4	Date:
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Date: 2026-10-08

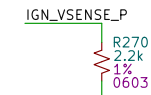
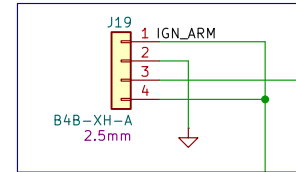
Rev: A

SIZE: A1	
KiCad E.D.A. 10.0.5	

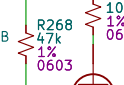
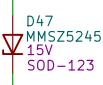
Id: 16/32

IGNITER SYSTEM

ARMING SWITCH
EXTERNAL LED ARMED
CONNECTOR



D33 clamps Q8
Vsg at -15V
IOUT MAX: 14.7A



Current Limit: 14.7A
Fault blanking: 22ms
Output rise time: 0.46V/ms

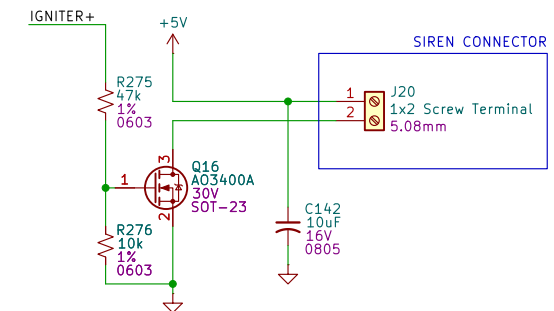
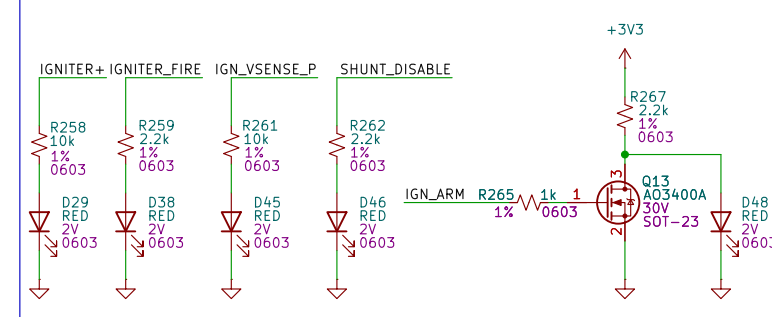
Continuity check:
1. ELOAD disabled, Q8 disarmed.
2. K1 shorts pins 1-3.
3. MCU checks continuity_sense net:
i. If HIGH, igniter not present
ii. If LOW, igniter present

sense @ 12V: 1.69V
sense @ 16V: 2.25V
sense @ 20V: 2.82V

TEST POINTS

TP41 TestPoint	IGN_VSENSE_P	TP67 TestPoint	IMON
TP56 TestPoint	IGNITER_FIRE	TP68 TestPoint	ITIM
TP61 TestPoint	EFUSE_OVLO	TP69 TestPoint	SENSE
TP62 TestPoint	SHUNT_DISABLE	TP70 TestPoint	+5V
TP63 TestPoint	IGNITER+	TP71 TestPoint	IGN_RLY-
TP64 TestPoint	DVDT		
TP65 TestPoint	PGOOD_IGNITER		
TP66 TestPoint	ILIM		

LEDs



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Sheet: /Igniter/
File: igniter.kicad_sch

Title: **IGNITER SYSTEM**

Size: A3 Date: 2026-10-08

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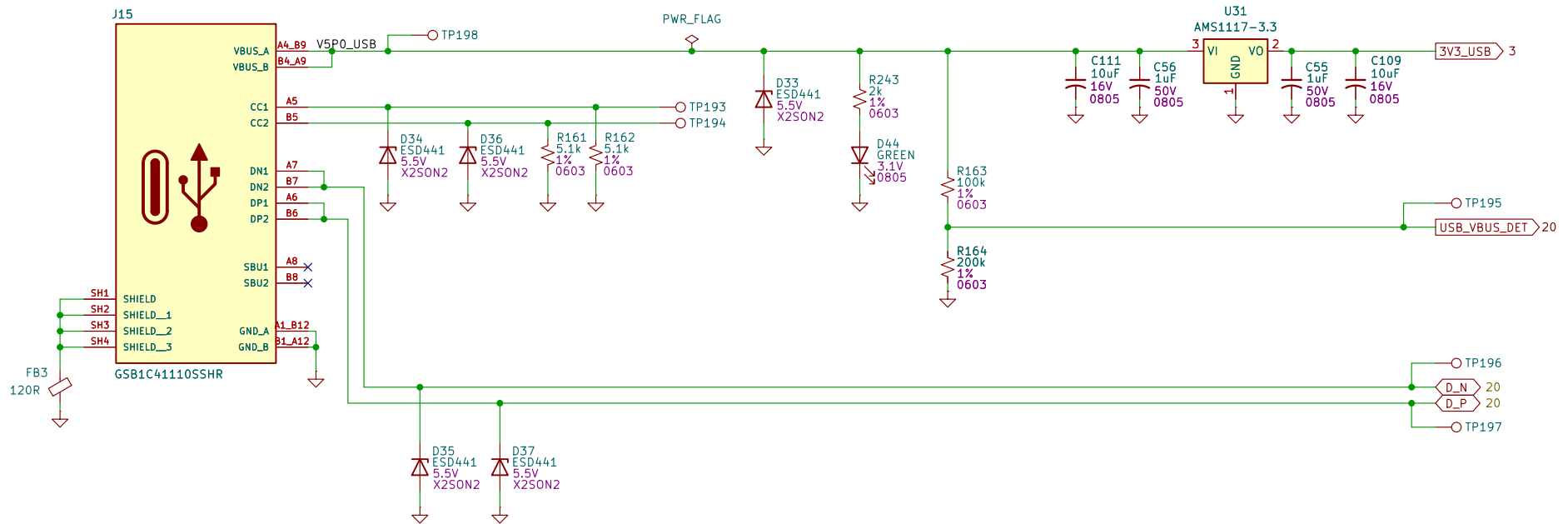
Rev: **A**

Id: 18/32



[illegible]

USB-C



Sheet: /USB-C/
File: usb-c.kicad_sch

Title:

Size: A4 Date: 2026-10-08

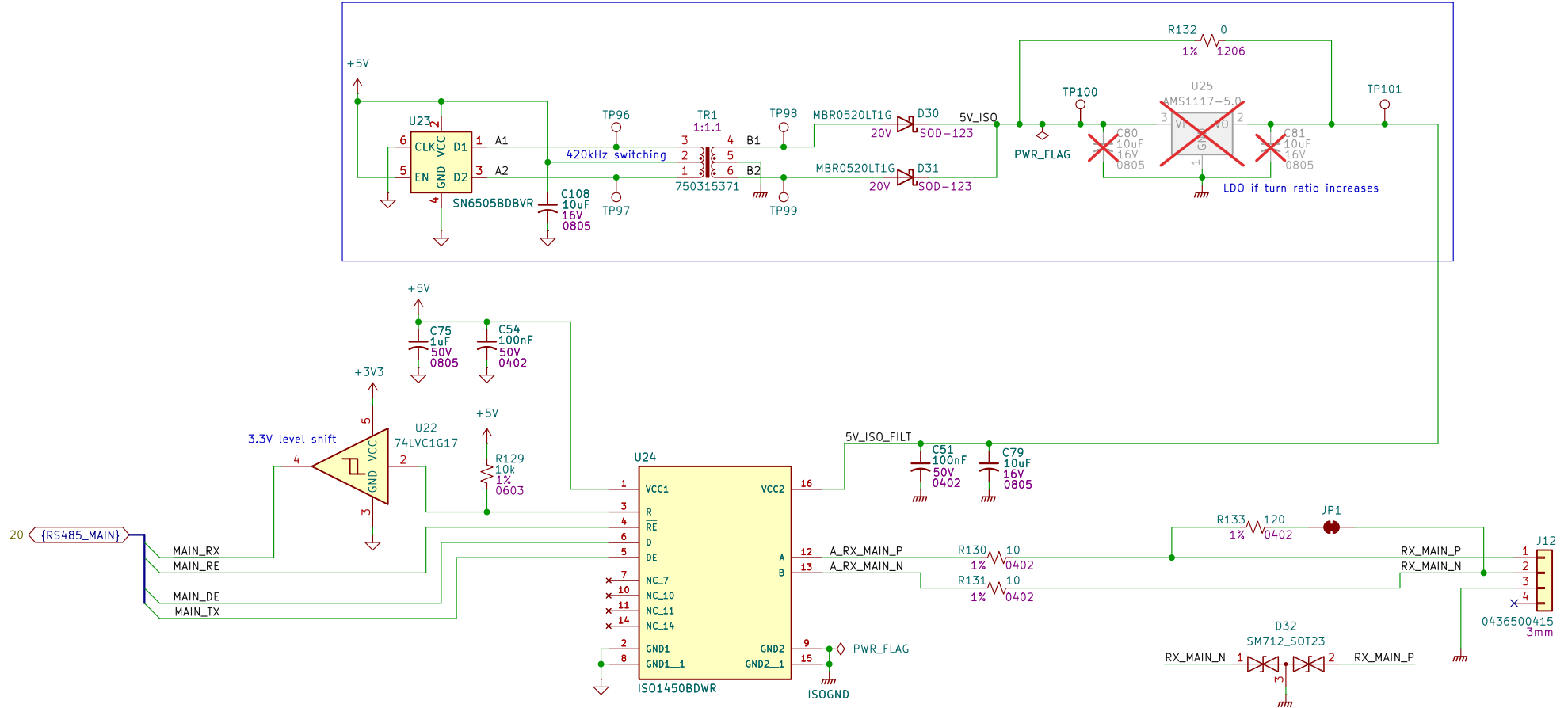
KiCad E.D.A. 10.0.5

Rev: A

Id: 22/32

RS-485

Galvanic Power Isolation



Sheet: /RS-485/
File: rs-485.kicad_sch

Title: RS-485

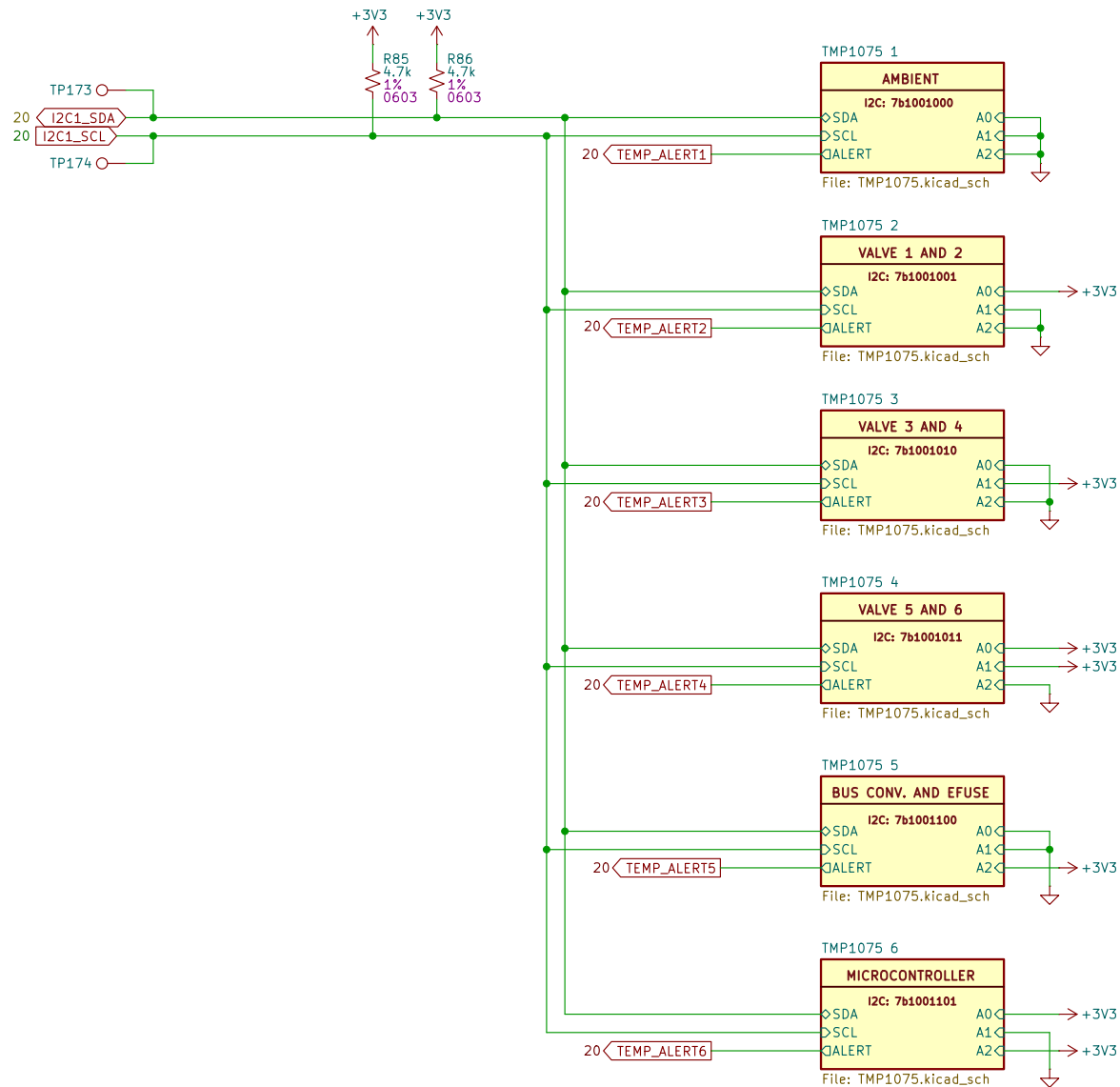
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KiCad E.D.A. 10.0.5

Rev: A

Id: 23/32

TEMPERATURE SENSORS

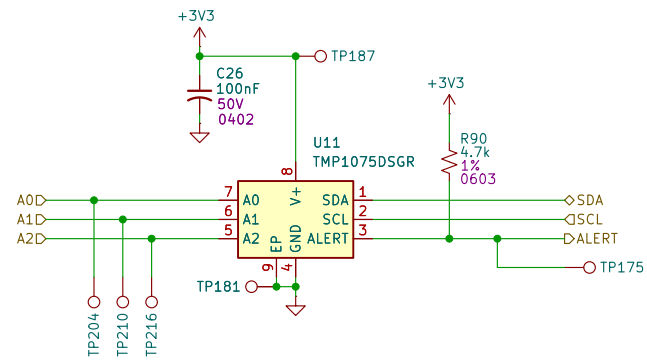




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Sheet: /Sensors/		Rev: A
File: sensors.kicad_sch		
Title: SENSORS		Id: 24/32
Size: A4	Date: 2026-10-08	
KiCad E.D.A. 10.0.5		

TMP1075



Sheet: /Sensors/TMP1075 1/
File: TMP1075.kicad_sch

Title: TMP1075

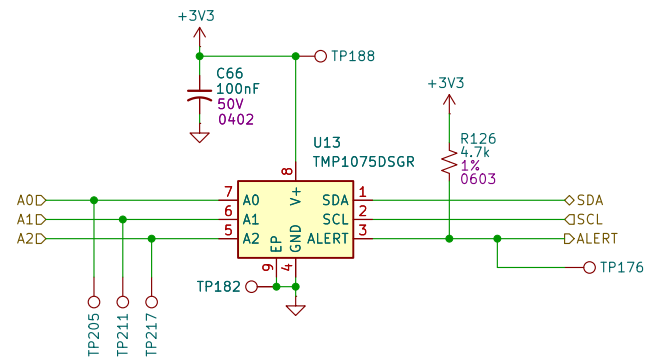
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KiCad E.D.A. 10.0.5

Rev: A

Id: 25/32

TMP1075



Sheet: /Sensors/TMP1075 2/
File: TMP1075.kicad_sch

Title: TMP1075

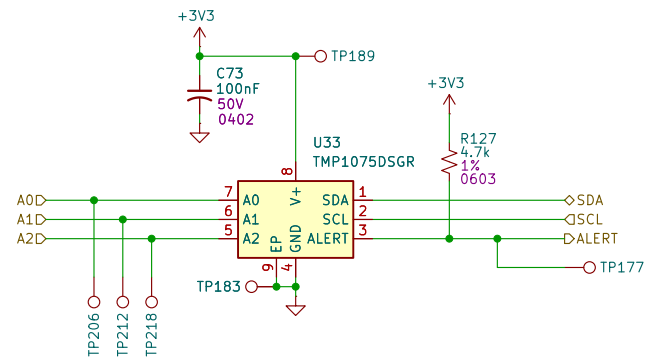
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KiCad E.D.A. 10.0.5

Rev: A

Id: 26/32

TMP1075



Sheet: /Sensors/TMP1075 3/
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Title: TMP1075

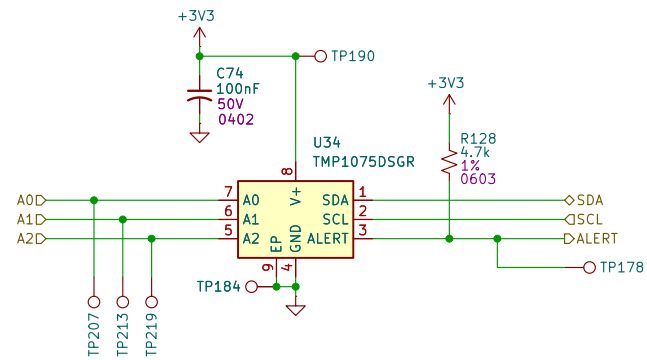
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KiCad E.D.A. 10.0.5

Rev: A

Id: 27/32

TMP1075



Sheet: /Sensors/TMP1075 4/
File: TMP1075.kicad_sch

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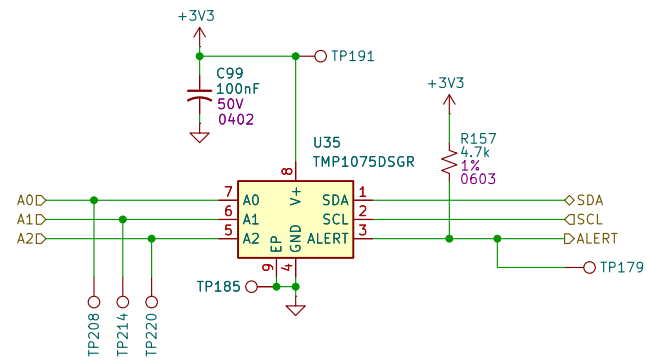
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 28/32

TMP1075



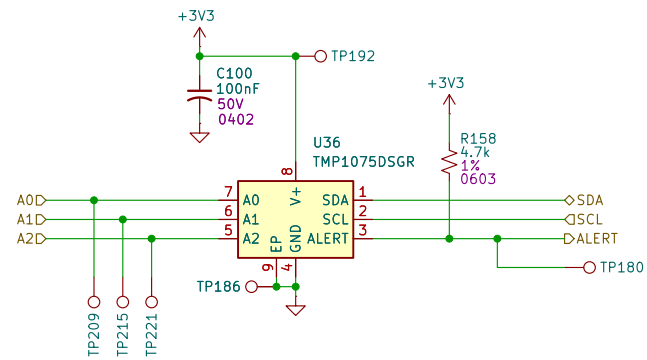
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Title: TMP1075

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

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Id: 29/32

TMP1075



Sheet: /Sensors/TMP1075 6/
File: TMP1075.kicad_sch

Title: TMP1075

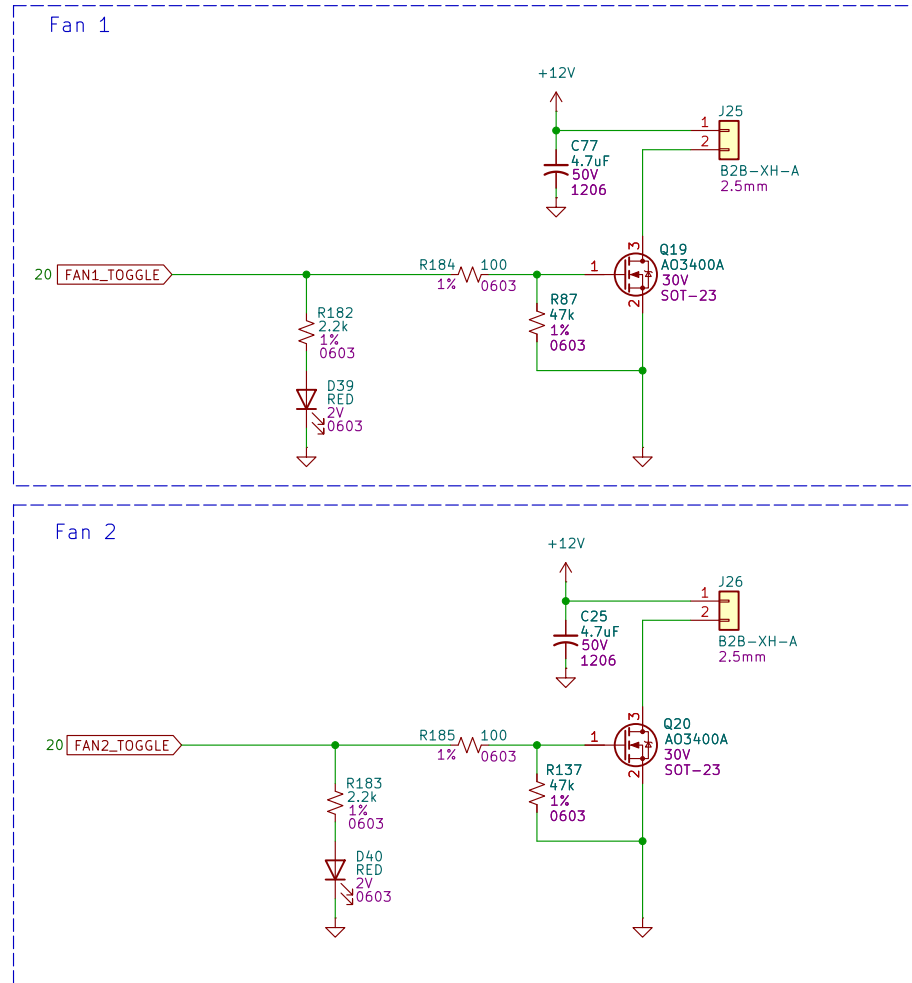
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 30/32

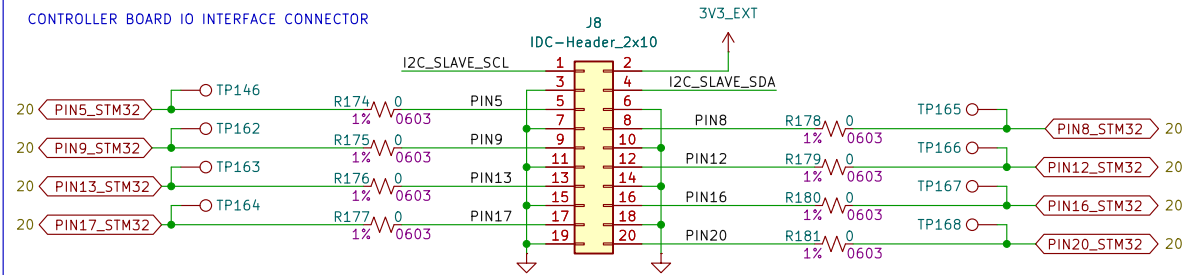
FANS



Sheet: /Fans/ File: fans.kicad_sch		
Title: Fans		
Size: A4	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.5		Id: 31/32

MECHANICAL

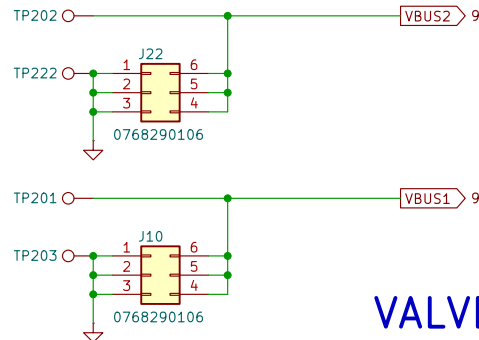
CONTROLLER BOARD IO INTERFACE CONNECTOR



SYSTEM POWER



VALVE POWER

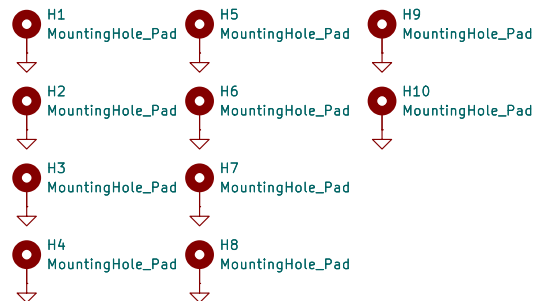


FID1
Fiducial
1mm

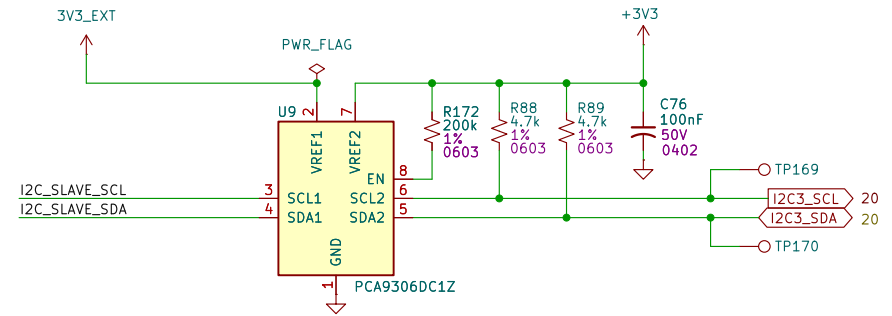
FID2
Fiducial
1mm

FID3
Fiducial
1mm

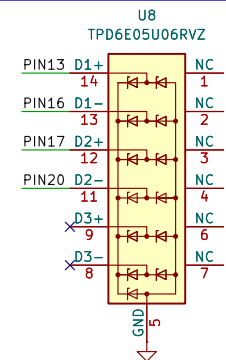
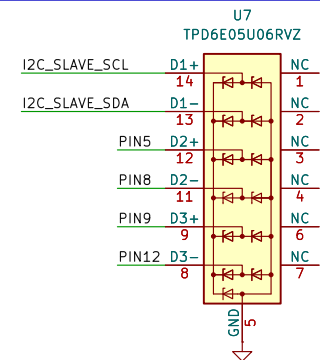
FIDUCIALS



MOUNTING HOLES



I2C BUFFER



ESD PROTECTION

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Sheet: /Connectors/
File: connectors.kicad_sch

Title: MECHANICAL

Size: A4
KiCad E.D.A. 10.0.5

Date: 2026-10-08

Rev: A
Id: 32/32